Assignment 1

for CS232/NetSys201 Fall 2017

Problem 1:

Suppose there is a 10Mbps microwave link between a geostationary satellite and its base station on Earth. Every minute the satellite takes a digital picture (of the earth) and sends it to the base station. Assume that propagation speed is $2.99x10^8 m/s$.

a) What is the propagation delay?
b) What is the bandwidth-delay product, $R * d_{prop}$
c) Let X be a the size of the photo. What is the minimum value for x that keeps the link continuously transmitting?
d) If the pictures are half of such size, what is the maximum interval of time between pictures that keeps the link busy?
e) What is the average queueing time if I take a burst of 100 pictures of size S? What is the maximum S such that I finish transmitting them before the next scheduled picture is taken
f) How wide is a bit over the channel?
g) What is the size of a picture such that it starts to be received before the transmitter stops sending it? And what is the maximum amount of bits that can be on the channel at the same time?

Problem 2:

(cont from P1) Let's now suppose that the satellite takes pictures every Y seconds, with Y distributed as an exponential random variable, with mean one minute.

f) What is the probability a picture will be taken in the next K seconds?
g) What is the average time at which the base station will have the next picture?

Problem 3:

Suppose N packets arrive simultaneously to a link in which no packets are currently being transmitted or queued. Each packet is of length L and the link has transmission rate R.

a) What is the average queueing delay for the N packets?
b) For each problem parameter, how can modify it (increase/decrease/uncorrelated) in order to decrease the average queueing time?
c) What is the maximum rate at which the link can accept packets if it doesn't have a queue?

Problem 4:

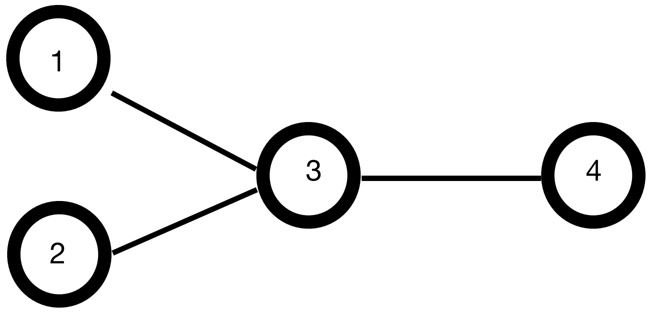
Consider a link between a Server and a Client that is the concatenation of N links, each of which has a packet loss probability p, and suppose that packet loss probabilities are independent for each link. What is the probability that a packet, sent from the server, is received at the client side? If a packet is lost in the path, the server "will know" and retransmit a new packet. How many times, on average, will the server have to send the packet over the link?

Problem 5:

Suppose that we have a network as in Figure 1. Routers 1,2,3,4 forward packets at fixed rates $R_1 = 0.5Mbps, R_2 = 2Mbps, R_3 = 2Mbps, R_4 = 3Mbps.$

Packets A, B and C have all same length L = 1 Mbps. A starts from router 1, B and C from router 2. Routers 1 and 2 start transmitting packets A, B simultaneously, and 2 sends B and C back to back.

A) find the end-to-end delay of packets A,B,C
B) Solve the problem for $R_3 = 1Mbps$



Problem 6:
A server receives packets from two different clients. The time between the generation of two consecutive packets at each client is exponentially distributed with parameter λ_1 for node 1, and λ_2 at node B. Call A the next packet that 1 sends, and B the next packet node 2 sends. The time the server takes to serve one packet is also exponentially distributed, with parameter μ. Assume no time is in the buffer at time 0.
A) find the probability packet A finishes transmission before the second packet B arrives.
B) Define N as the number of packets in the buffer after packet B arrives. Find E[N].
C) compute the average permanence time of packet B in the buffer (waiting time plus transmission time).
D) define T as the time when the last bit of packet B is transmitted by the router. Find E[T].

Problem 7: (extra)
Experiment with P19 in the Kurose Ross. Report any interesting result here.